Hackathon Finder

A skill-based, explainable search tool for student hackathons

Prototype Report

Submitted by	_____
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Live prototype	https://claude.ai/artifact/1GZsb28N1Jo4F93NDEk1cQ

Note: The prototype uses sample data. It demonstrates the interface, filtering and matching logic, not a live feed of real events.

1. Problem statement

Students who want to join hackathons face three practical problems. Events are scattered across many websites, social media posts and college groups. Each listing mixes eligibility rules, deadlines and formats in different ways. And there is no easy way to tell which events actually suit a student's skills, so many students miss deadlines or join events that do not fit them.

2. Objective

Build a single place where a student can describe their skills once and immediately see hackathons ranked by fit, with a clear reason for each ranking.

3. Proposed solution

Hackathon Finder is a web application with a filter panel and a ranked results list. The student selects skills, picks online or in-person events, and optionally narrows to Delhi NCR. Each result shows a match percentage, the days left to apply, the prize pool, eligibility and a short explanation of why it was ranked there.

4. Key features

- **Skill-based matching:** every hackathon has theme tags; the student's skills are compared against them.
- **Explainable score:** the match percentage is never a black box. Matched themes are highlighted and a plain sentence explains the score.
- **Filters:** search by name or theme, online or in-person, Delhi NCR only, saved only.
- **Sorting:** best match, closest deadline or highest prize pool.
- **Deadline urgency:** events closing within 7 days are marked so they are not missed.
- **Save list:** students can shortlist events and view only those.
- **Responsive and accessible:** works on phones, supports keyboard use and light or dark themes.

5. How the match score works

The score is deliberately simple so it can be explained to any user and examined by an evaluator:

Match % = (themes of the hackathon that the student has selected) divided by (total themes of the hackathon) x 100

Example: a student selects Python, AI/ML and Web.

Hackathon	Themes	Matched	Score
Agent Builders Online Cup	AI/ML, Automation, Python	AI/ML, Python	67%
AI for Bharat Build-a-thon	AI/ML, Python, Data	AI/ML, Python	67%
Delhi NCR Student Hack Night	Web, JavaScript, Open Source	Web	33%
FinTech Automation Challenge	FinTech, Automation, JavaScript, Cloud	None	0%

Ties are broken by the closest deadline, so the most urgent equal match appears first.

6. System design

The prototype is a single-page web application that runs entirely in the browser. The production version would add a backend and database, as shown below.

Layer	Prototype (built)	Production version (planned)
Interface	HTML, CSS, JavaScript; responsive layout	React front end with the same components
Data	Sample list inside the page; deadlines generated relative to today	Database table of events (Supabase or PostgreSQL)
Logic	Filtering, scoring and sorting in JavaScript	Same scoring on the server, with API endpoints
Data collection	Manual sample data	Scheduled scraper or API pulls from hackathon platforms, plus organiser submission form
User accounts	None (state kept in memory)	Login, saved skills and saved events per user
Alerts	None	Email or WhatsApp reminder before a deadline

7. Technology used

- HTML5 and CSS3 for structure and responsive styling (CSS variables for light and dark themes)
- Vanilla JavaScript for state, filtering, scoring and rendering
- Google Fonts (Bricolage Grotesque, IBM Plex Sans)
- Hosted as a static page; no server is required for the prototype

8. Limitations of the prototype

- All events are sample data and registration links are not connected.
- Matching uses exact tag overlap. It does not yet understand related skills (for example, Machine Learning and AI/ML).
- Saved events and skills reset when the page is closed.

9. Future scope

- Collect real events automatically from public hackathon platforms and college pages.
- Weight skills by level (beginner, intermediate, advanced) and add team-size and travel-distance preferences.
- Use text embeddings so that related skills and project ideas match, while keeping the explanation visible.
- Reminders before deadlines, and a team-finder for students who need teammates.

10. Conclusion

Hackathon Finder shows that a small, transparent matching rule combined with good filters can save students time and help them apply to events that suit them. The prototype demonstrates the full user flow; the next step is connecting it to real, regularly updated event data.